

Koa Shen

Mechanical Engineering Intern Candidate · Robotic Mechanisms · Actuation · Prototyping

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EDUCATION

University of California, Santa Barbara

Expected June 2028

B.S. Mechanical Engineering · 3.97 major / 3.94 overall GPA · Dean's Honors every quarter

Santa Barbara, CA

- Coursework: Dynamics, Mechanics of Materials, Circuits. Current: Dynamical Systems and Machine Learning.

EXPERIENCE

Mechanical Engineering Intern

Summer 2026 – Present (extended to part-time remote)

SLAC National Accelerator Laboratory — LCLS

Menlo Park, CA

- Designed precision motion assemblies in Solid Edge and Teamcenter, applying GD&T and statics/dynamics checks to mounts and stages.
- Used Python/Drake kinematics and collision checks to identify interference and guide mechanical changes to packaging and alignment.

Fabrication Team Lead

Fall 2024 – Present

Gaucho Racing (FSAE EV) — UC Santa Barbara

Santa Barbara, CA

- Led a 15-person team from design through fabrication and installation of driver-safety hardware; passed FSAE technical inspection.
- CNC-machined 33 suspension components in 4130 steel and 7075-T6 aluminum; held bearing interfaces within 0.0005 in.

Maintenance Technician

Summer 2025

Exploratorium

San Francisco, CA

- Reverse-engineered and repaired mechanisms across 100+ interactive exhibits using manual machining, welding, and fabrication tools.
- Redesigned a cracking acrylic enclosure interface with a load-spreading bezel, avoiding \$3,000+ in replacement costs; documented CAD for future repair.

SELECTED PROJECTS

Compact Cycloidal Reducer & Stepper Package

Summer 2026 – Present

- Designed and built a compact 15:1 cycloidal reducer package for a NEMA 17 stepper, integrating a TMC2209 driver and AS5600 encoder.
- Bench-validated encoder feedback, missed-step detection, and automatic position correction in custom C++ firmware; printed reducer parts in PETG/ABS.

GR26 CNC Wheel Assemblies

Winter 2025 – Spring 2026

- CNC-machined 33 precision suspension parts on Haas mills and lathes; held critical bearing fits within 0.0005 in and standardized five spindles.

FSAE Steering Wheel Development

Fall 2024 – Spring 2026

- Integrated buttons, potentiometers, display, and quick-release into a driver interface; FEA-guided iteration reduced mass from 5 lb to 1.7 lb.

TECHNICAL SKILLS

Mechanical Design: Solid Edge, SolidWorks, Mechanism design, GD&T, Teamcenter PDM, Tolerance-aware design

Prototyping & Manufacturing: CNC/manual machining, 3D printing (PETG/ABS), Mastercam, FEA, Assembly and functional testing

Mechatronics & Controls: C/C++, Python, Arduino, TMC2209 stepper control, AS5600 encoder feedback, PlatformIO